

Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-8336
Project Name	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how EduGenie performs under normal usage and multiple user requests. It checks response time, error rate, CPU usage, and memory usage to ensure that the application provides a fast and reliable learning experience.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Chrome DevTools / Lighthouse
Type of Testing	Response Time Testing, Load Testing
Target Module / API	AI Chat / Google Gemini API, Quiz Generation
Test Environment	Localhost / Development Server
Test Date	01 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Student sends an AI doubt/question	1	60	Response received within a few seconds
2	Multiple students request AI explanations	5	60	System responds without errors
3	Students generate quizzes simultaneously	5	60	Quiz generation works correctly
4	Multiple users access dashboard and learning features	10	60	Application remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Good response time
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Acceptable under load
3	Throughput (Req/sec)	≥ 5	6	Pass	Handles requests properly
4	Error Rate	< 1%	0%	Pass	No major errors
5	CPU Utilization	< 80%	45%	Pass	CPU usage is normal
6	Memory Utilization	< 80%	62%	Pass	Memory usage is acceptable

Step 4: Observations & Analysis

Key Findings

EduGenie provides a responsive interface for normal student activities such as asking questions, generating explanations, and creating quizzes. The main response time depends on the Google Gemini API and network connection.

Bottlenecks Identified

nAI response generation and network latency may increase the response time when multiple requests are made at the same time.

Optimization Steps Taken

API requests were kept simple and unnecessary repeated requests were avoided. The interface was designed to provide clear feedback while waiting for AI responses.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

localhost:5173/dashboard

EduGenie
GEMINI AI POWERED

Dashboard

AI Tutor

Explain a Topic

Generate Notes

Generate Quiz

Evaluate Answer

Study Plan

Learning History

Progress

Profile

Settings

 **Kamali Senthil ku...**
Computer Science Student

Welcome back, Kamali! 🍌
Continue your learning journey and improve your skills.



Welcome back, Kamali! 🍌
Here's your learning overview.

34
Quizzes Completed
vs last month

88%
Average Score
Consistent high grade

42
Topics Studied
5 new this week

12 Days
Study Streak
Personal record: 18

Quick Actions
Choose an AI learning capability to start studying

AI Tutor
Ask questions and learn with your personal AI tutor.
Open →

Explain a Topic
Understand difficult concepts with simple explanations.
Open →

Generate Notes
Create structured study notes instantly.
Open →

Generate Quiz
Practice with AI-generated questions.
Open →

Evaluate Answer
Get AI feedback on your answers.

Study Plan
Create a personalized learning schedule.

Learning History
Review your previous learning activities.

Progress
Track your learning performance and growth.

27°C Clear



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01-10-2026